COOLING ENVIRONMENTS &
LIQUID COOLING

Door Heat Exchanger Sub-Project Updates

Jabari George *Consulting Mechanical Engineer, LGTS*

Sub-Project Lead, Door Heat Exchanger

Workstream Leads

Rajesh Kasukurthy, AMD · Adrian Zebrowski, Meta · Joe Thomas, Stulz

Ashok Yadav, Vertiv · Dominik Sporna, Valeo · Mohamad Hnayno, OVHCloud

Sub-Project Overview

Goals

Contribute documentation and products to develop the wider community of solutions

Lower the barriers to liquid cooling adoption

Harmonize with partner sub-projects/projects on definitions, references and standards

In-Scope Activities

Operating conditions and parameters

Metrology of heat extraction performance

Requirements for different rack architectures: Open Rack, EIA310, etc.

Solutions for data centers with facility water

Solutions for air-cooled data centers

3
Active Workstreams

5+
Community Presentations

15+
Contributing Organizations

Community Presentations

Informal presentations by industry professionals discussing current topics

September 2025 — **TESS Energy**

Waste Heat & Thermal Energy Storage Applications

March 2026 — **AVTRON**

Liquid-Cooled Load Banks in Datacenter Commissioning

June 2026 — **Kuhltherm**

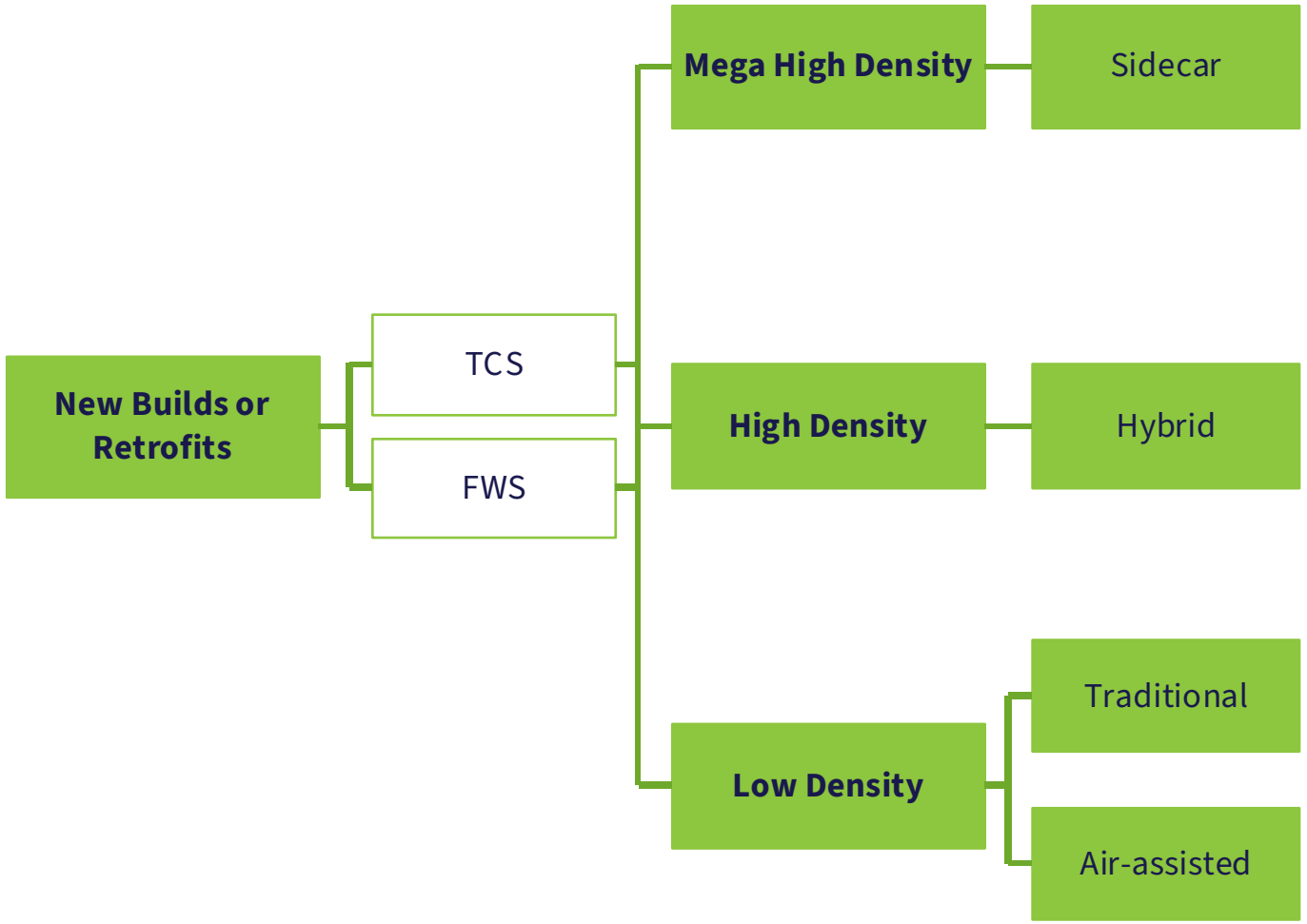
Modular Door HX Based on Automotive Radiator Technology

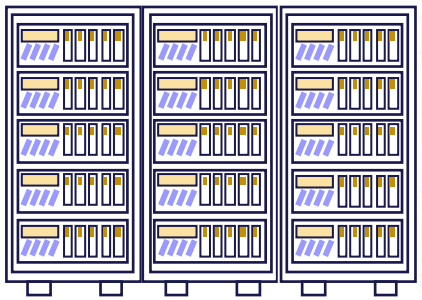
July 2026 — **Asperitas**

OCP TCO Effort and Its Expansion to Cover Door HX

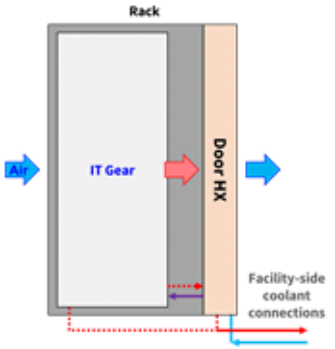
Topics arise from sub-project calls. Workstreams are created based on community interest. Future presentations are being arranged.

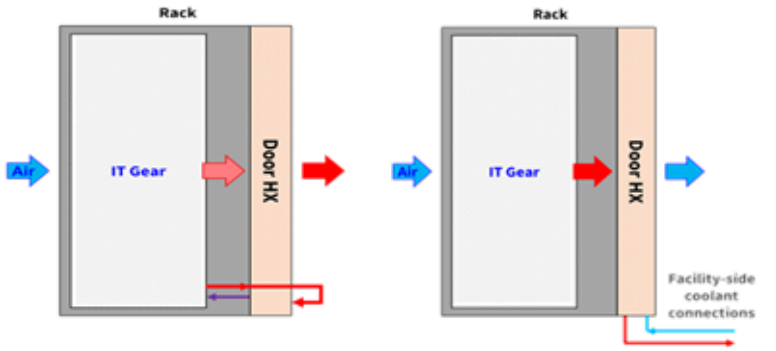
Future of Door Heat Exchangers



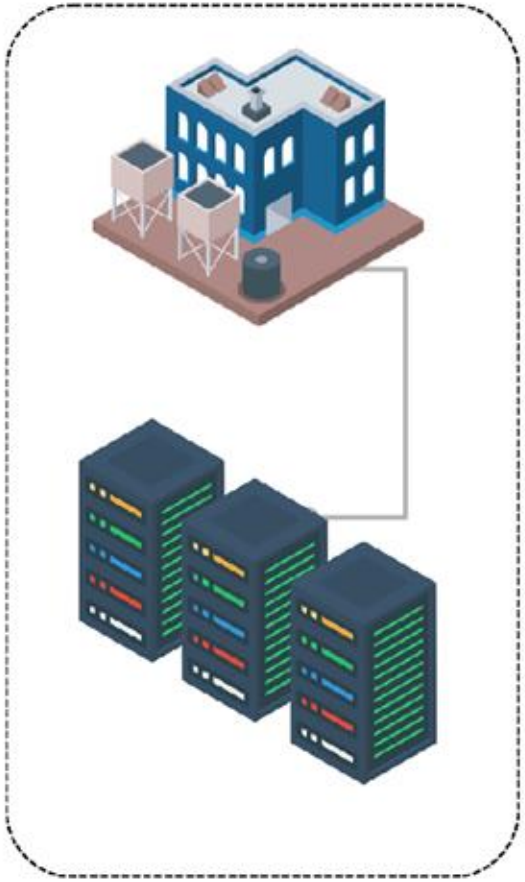




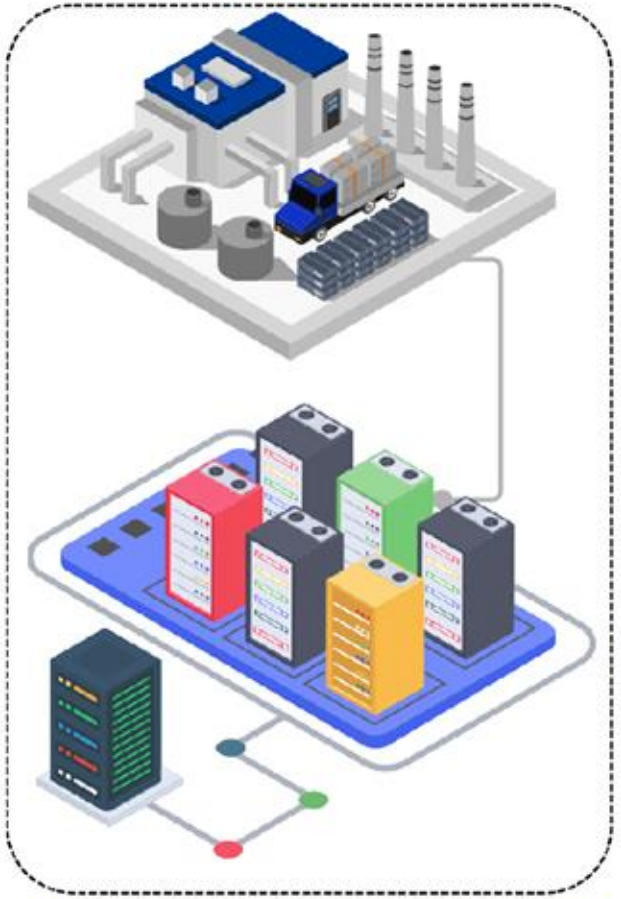




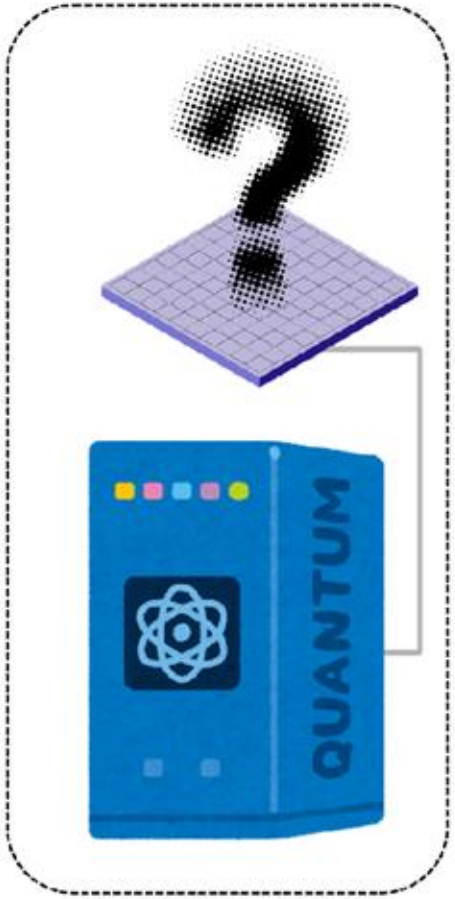
Future of Door Heat Exchangers



Low Density Facility



High Density Facility



Future?

Future of Door Heat Exchangers

Alternative metals and alloys



Alternative form factors



Alternative coolants & refrigerants



Aluminum Heat Exchangers

AALC Sidecar Solutions

Heat Exchanger Specification

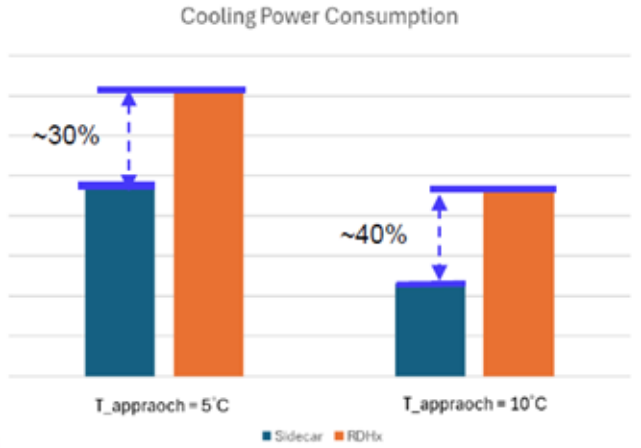
Workstream: Mega AALC for High Power Density AI Data Centers

IN PROGRESS

AALC combines direct-to-chip liquid cooling for CPUs/GPUs with air cooling for lower heat flux components, enabling efficient hybrid cooling for next-generation AI racks.

Key Sidecar Design Guidelines

- Dimensions aligned with IT rack standards and ORV3; form factors in Single Rack Space (SRS) units
- Critical components (pumps, fans, PSU) require redundancy and hot-swap capabilities
- Single-point-of-failure components (heat exchangers, manifolds) need careful design
- Support for both single-phase and two-phase coolants with appropriate material compatibility



Sidecar vs. RDHx Performance

Lower Power

Sidecar has significantly lower power consumption; delta increases with higher rack densities

Greater Efficiency

Sidecar exhibits greater cooling efficiency at scale with higher coolant flow rates

Higher Capacity

Sidecar achieves higher cooling capacity, with delta increasing at higher flow rates

Mega AALC Sidecars: Progress & Next Steps

Completed

- Whitepaper *"AALC Sidecar Solution for Enabling High Power Density Hardware"* submitted to OCP Contribution Portal.
- DC vs. EC fan analysis completed; efficiency and CFM/watt metrics added
- HX modeling underway: aluminum and copper configurations being simulated
- Draft whitepaper development initiated; test methodology metrics defined
- Proposal document reviewed and updated with community feedback
- New contributors joined for cabinet/logistics and fluid compatibility

Next Steps

- Add radial fans and efficiency metrics to the fan comparison
- Create optimized HX design comparison table
- All subheading owners contribute to the draft whitepaper
- Advance control strategy framework
- Continue mega sidecar design guidelines for 500kW+ AI racks

Mega Sidecar Scale

- Single-phase: up to 10+ rack widths for 1MW cooling
- Two-phase coolant reduces sidecar width significantly

Key Design Focus Areas

- Micro-level filtration for advanced cold plates
- High availability with hot-swap components
- Logistics planning for multi-rack deployment

AI Rack Trends

- Rack power projected to exceed 1MW by 2030
- Liquid cooling adoption trending toward 98%
- Existing DC infrastructure needs hybrid solutions

Workstream: Aluminum Heat Exchanger Integration

IN PROGRESS

Sustainable Materials

Copper to aluminum: cost savings, weight reduction, comparable thermal performance.

Corrosion Mitigation

Advanced coatings, sacrificial layers, adherence to VDI 2035 and OCP guidelines.

Infrastructure Integration

Prototype designed for compatibility with existing OVHCloud rack systems.

Specialized Design

Folded tube design with sacrificial layers and robust headers/side plates.

Thermal Performance: Copper vs. Aluminum Heat Exchangers

Parameter	Copper Tube HX	Aluminum Brazed HX	Difference
Cooling Capacity (W)	5,360	5,650	+5.4%
Water-side ΔP (bar)	0.36	0.12	−67%
Weight Unfilled (kg)	16.8	8.0	−52%

Aluminum HX: Corrosion Testing & Timeline

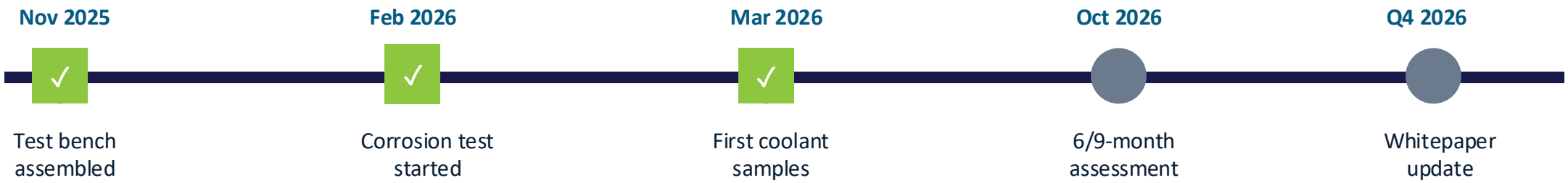
Test 1: Component Focus (Valeo)

- 6 aluminum radiators under test in data center fluid conditions
- Key fluid parameters analysis: pH, conductivity, ion release
- First data analysed; periodical metallurgical analysis continuing
- Fluid based on ASHRAE & OVHCloud inputs
- Samples analyzed following ASTM and automotive expertise

Test 2: System Focus (OVHCloud)

- Closed-loop hydraulic system with aluminum HX, water blocks, manifolds
- Continuous monitoring: flow rate, pressure, temperature, pH, conductivity
- Heater cycles 3200 W elements 5 min ON / 5 min OFF at 50 °C
- Testing coupons at 1-, 3-, and 6-month intervals
- Test bench running since February 2026

Testing Timeline



Door HX Specification Rev. 1.0

REV 0.5 IN REVIEW

Launched January 2026. Rev 0.2 comments are consolidated and the Rev 0.5 candidate is going out for community feedback, on track toward a Rev 1.0 specification.

Specification Scope

- Performance requirements and testing standards for Door Heat Exchangers
- Compatibility requirements for Open Rack and EIA310 architectures
- Operating conditions, parameters, and environmental considerations
- Metrology standards for heat extraction performance measurement
- Integration guidelines for facility water and air-cooled data centers

Current Participants

- Stulz, Vertiv and Chatsworth are contributing to this specification.
- If your organization manufactures OCP liquid cooling equipment, we encourage you to get involved.

Roadmap

- Rev 0.5 – Community Review (complete)
- Rev 0.8 – Review Draft
- Rev 1.0 – Formal Release (Q4 2026)

Door Heat Exchanger Sub-Project Contributors

Workstream leads driving the sub-project's technical output



Rajesh Kasukurthy

AMD
Thermal Engineer

Mega AALC Sidecars

AZ

Adrian Zebrowski

Meta
Hardware Thermal Engineer

Mega AALC Sidecars



Joe Thomas

Stulz
Director Consultant Management EMEA

Door HX Specification



Ashok Yadav

Vertiv
Global Offering Manager HDLC

Door HX Specification



Dominik Sporna

Valeo
Customer Chief Engineer

Aluminum HX Integration



Mohamad Hnayno

OVHCloud
Design Solutions Team Leader

Aluminum HX Integration

Get Involved!

Improve the performance, efficiency, and sustainability of Door Heat Exchangers and Sidecars.

How to Participate

- Monthly calls on first Thursday at 9 AM PST
- Join via QR codes
- Contribute to active workstreams
- Present on relevant topics during community presentations

Resources

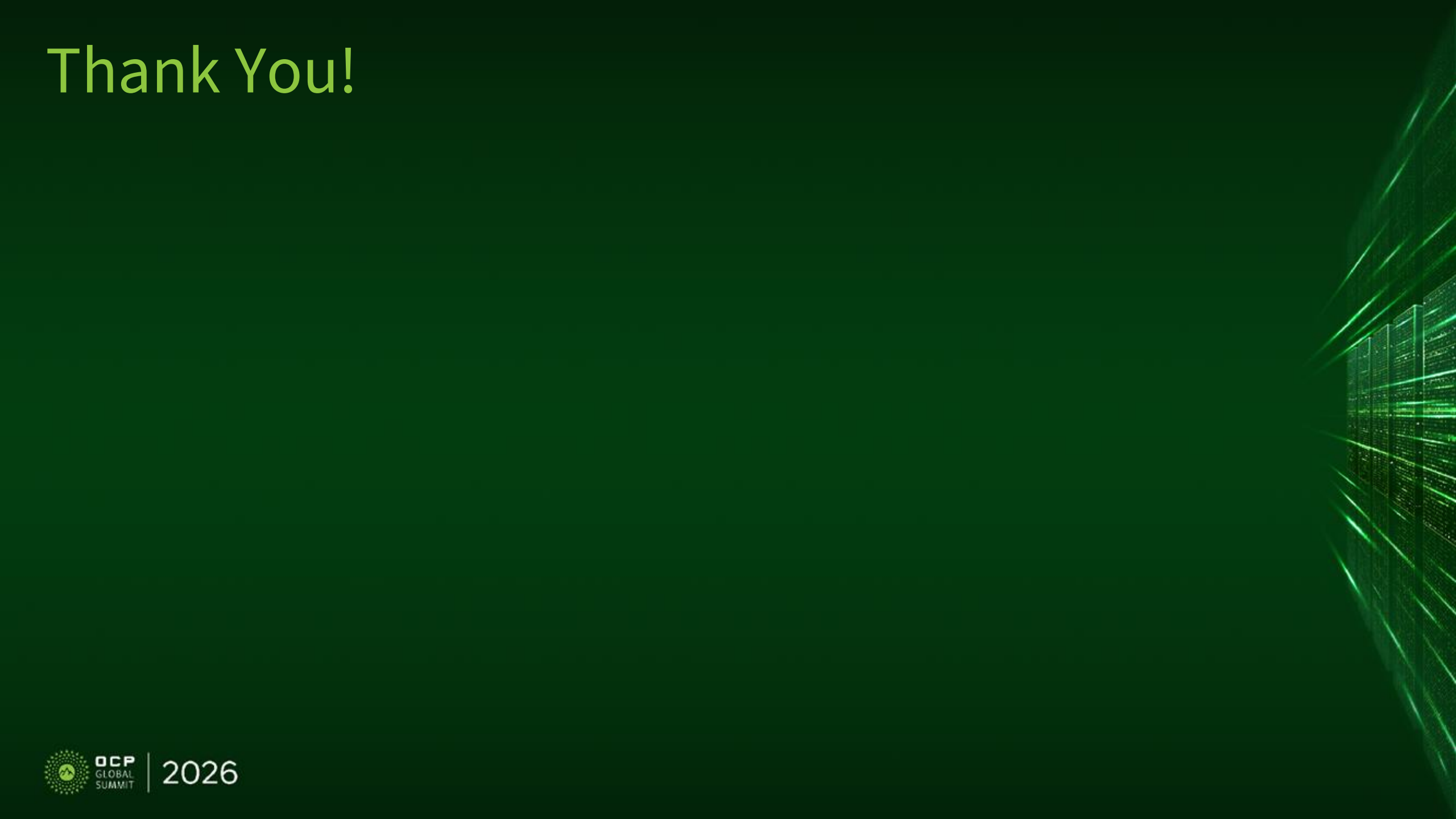
Mailing List

ocp-all.groups.io/g/OCP-ACS-Door-Heat-Exchanger

Project Wiki

opencompute.org/wiki/Cooling_Environments_Advanced_Cooling_Solutions_Door_Heat_Exchanger

Thank You!



Scaling Aluminum Heat Exchangers in Data Centers:
Performance, Corrosion , PUE validation updates

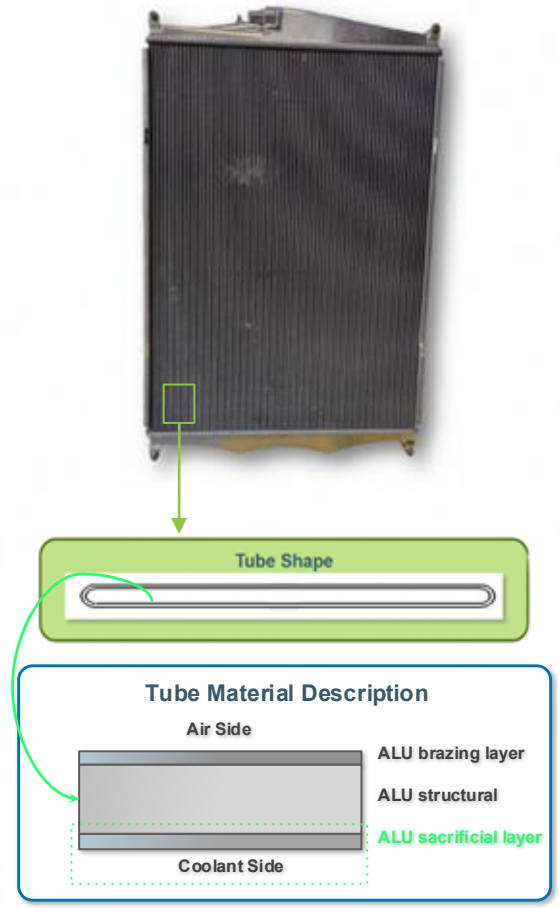
Mohamad HNAYNO PhD, *Design Solutions Team Leader , OVH CLOUD*

Dominik SPORNA *Customer Chief Engineer, VALEO*

Aluminum Heat Exchangers

CONSTRUCTION REMINDER

- Aluminium brazed core technology
- Based on strict automotive standards
- Variants with reinforced pressure resistance (> 300psi)
- Cooling Capacity up to 38% vs Copper RAD in ISO conditions
- Power density + 65% vs Copper Coil
- High scale yearly production (~ 30mln/y Globally)



Sacrificial layer - thin inner tube material layer with unique chemical composition that provides cathodic corrosion protection

Alu vs Copper Radiators comparison

Performance updates





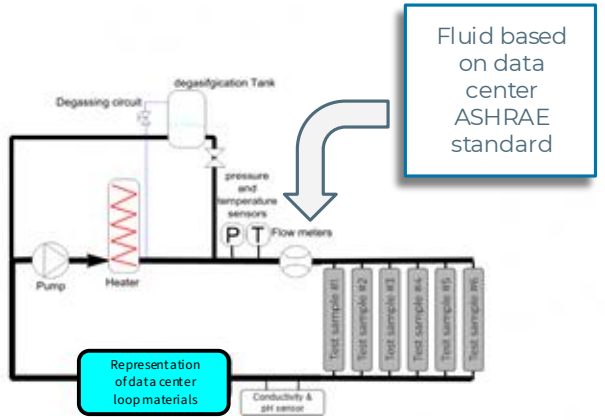
Parameters	Copper Radiator 48U	Aluminum Radiator (Valeo technology) 48U	Difference
Cooling Capacity [kW]	23.1	**32**	**+38.5%**
Dry Weight	35kg	**24kg**	**-31%**
Wet weight (with coolant)	43kg	**36 kg**	**-16%**
Power density (Cooling Capacity/Wet weight)	0.54 kW/kg	0.89 kW/kg	**+65%**
Air-side pressure drop [Pa]	62	72	+16%
Coolant-side pressure drop [kPa]	26	21	**-17%**

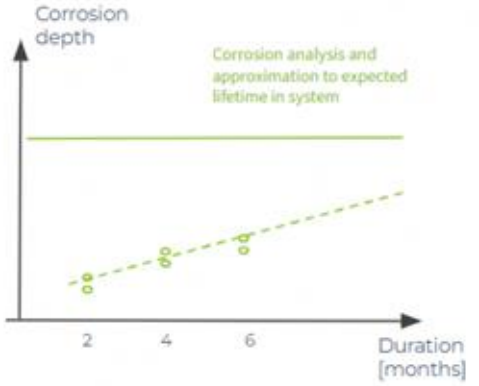
TESTING COMPLETED IN ISO CONDITIONS

Aluminum Heat Exchangers

COMPONENT Corrosion testing updates

- **CORROSION TESTING STATUS @ OVH**
  - 1 bench running since Q1 2026
  - See next page
- **CORROSION TESTING STATUS @ VALEO**
  - 2 accelerated factor benches started in Q2 2026
  - 2 additional benches planned in Q4 2026
  - Benches run in the environment representing data center loop materials
  - Testing with increased severity to accelerate testing duration while maintaining the expected lifetime
- **MEDIUMs under analysis**
  - Water based solution - Medium pH
  - Water based solution - High pH
  - PG25





RELIABLE SOLUTION UNDER DEVELOPMENT
TO PROVIDE LIFETIME OPERATION FOR ALL 3 TYPES OF MEDIUM

Aluminum Heat Exchangers

System testing - BENCH no.1 (OVH)



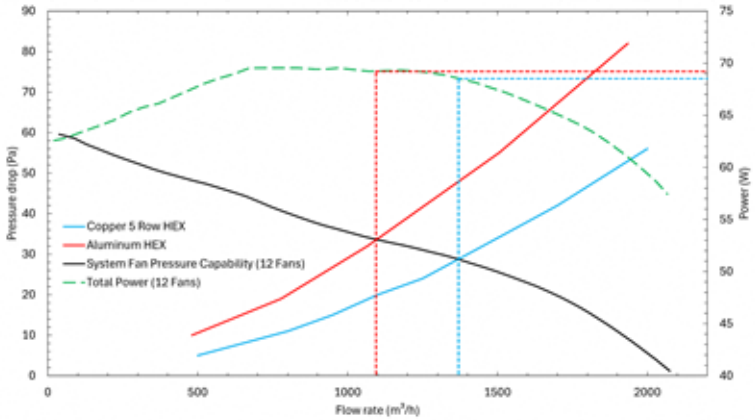
Continuous bench monitoring -reading example

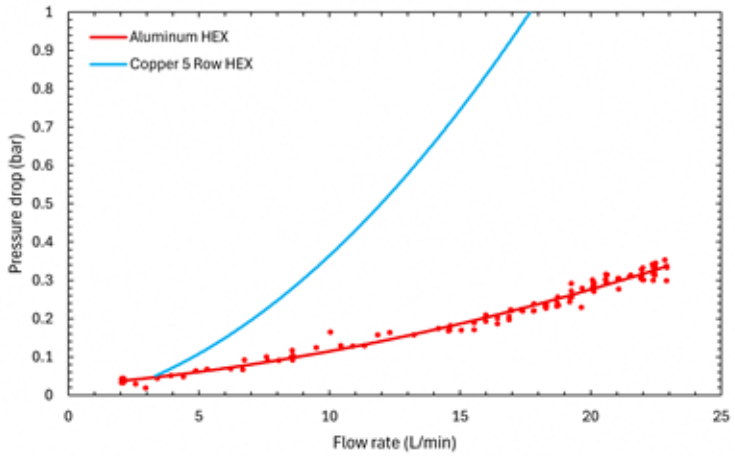
Parameter	Sample Day 1	Sample Day 1+ 4 Months
pH	8,56	7,7
Conductivity	1122 µS/cm	1700 µS/cm
Aluminium	< 0,1 mg/l	0,16 mg/l
Copper	< 0,01 mg/l	0,015 µg/l
Sodium	218 mg/l	410 mg/l
Potassium	5 mg/l	31 mg/l
Manganese	62 mg/l	508 mg/l
Carbone organique dissous (COD)	3.7 mg/l	2200 mg/l
Residual after filtration	1600 mg/l E/L	2900 mg/l E/L

Water quality check Day 1 + Day 120

Aluminum Heat Exchangers

10kW Rack cooling PPUE





	AC/WC Copper HEX (W)	AC/WC Alu HEX (W)
RDHX Fans power consumption	204	207
In rack CDU(x2) power consumption	240 (HD)	120 (HD)
Total cooling power consumption per rack (48s)	444	327
Cooling power consumption per U	9,25	6,8
Rack PPUE (10kW rack)	1.044	1.032

Cooling power consumption reduced by 25%

Aluminum Heat Exchangers

NEXT STEPS

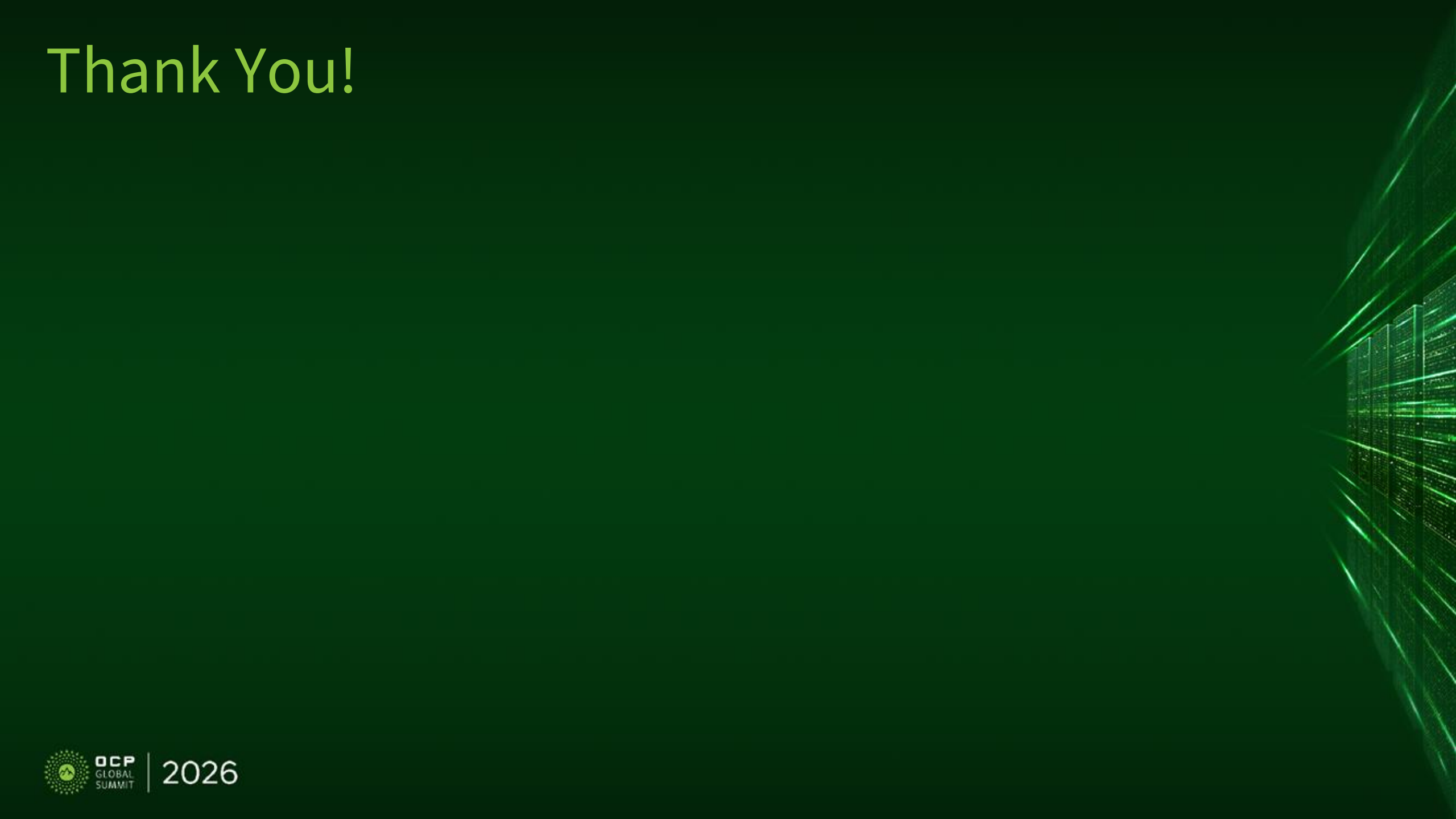
- Completion of all 5 scheduled testing by Q2 27
- Water sample analysis results in Q4 26 and Q1 27
- Material coupons analysis
- Post testing radiators analysis
- Fluid parameters recommendations
- Complete Aluminium Radiators Lifetime assessment

Call to Action

- **Problem to solve**
  - Create Guidelines for Aluminum Heat Exchanger Deployments for OCP
- **Weekly workstream calls on Tuesdays at 8 AM PST**
- **White Paper update available in S1 2026**
- **Where to find additional information (URL links)**
  - Door Heat Exchanger Wiki: https://www.opencompute.org/wiki/Cooling_Environments_Advanced_Cooling_Solutions_Door_Heat_Exchanger
  - Door Heat Exchanger Mailing List https://ocp-all.groups.io/g/OCP-ACS-Door-Heat-Exchanger



Thank You!



Mega sidecar AALC to enable AI hardware in legacy data center infrastructure

Rajesh Kasukurthy *Thermal Engineer , AMD*

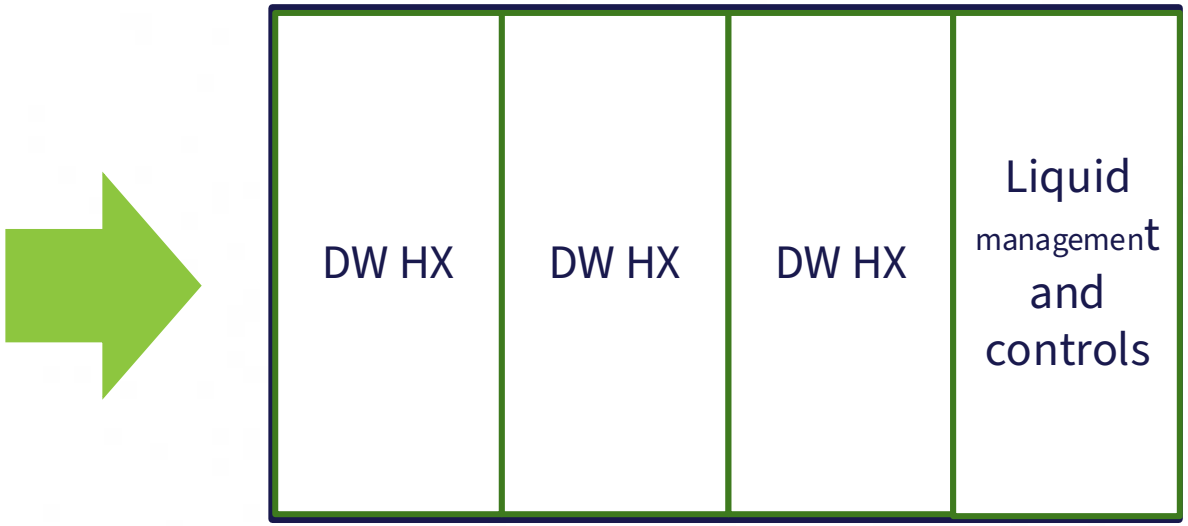
Adrian Zebrowski *Thermal Engineer, Meta*

Mega sidecar

Mega side can is designed for up to 1 MW of rack power

The building block of the side can is OCP double wide rack form factor heat exchanger (DWHX)





AI Chip Rack TDP ~ 500 kW

Mega sidecar HX performance

Assumptions: Double wide rack form factor rack

Boundary conditions:

Air in Temp: 30C, Air out: 40C

Liquid in: 50C, Liquid out to server: 40C

Airflow: 18000 CFM

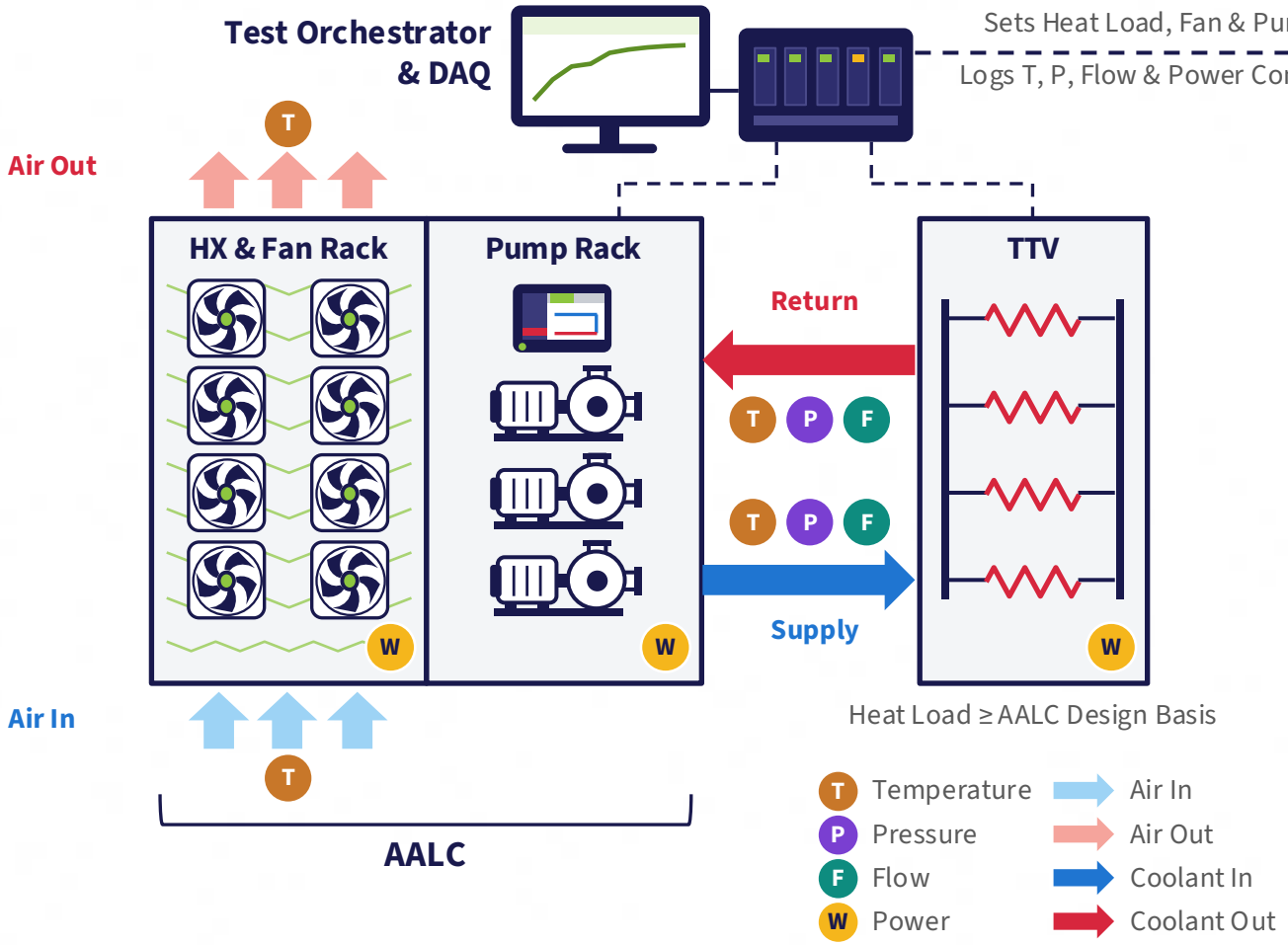
Liquid flow: 1.5 LPM/kW

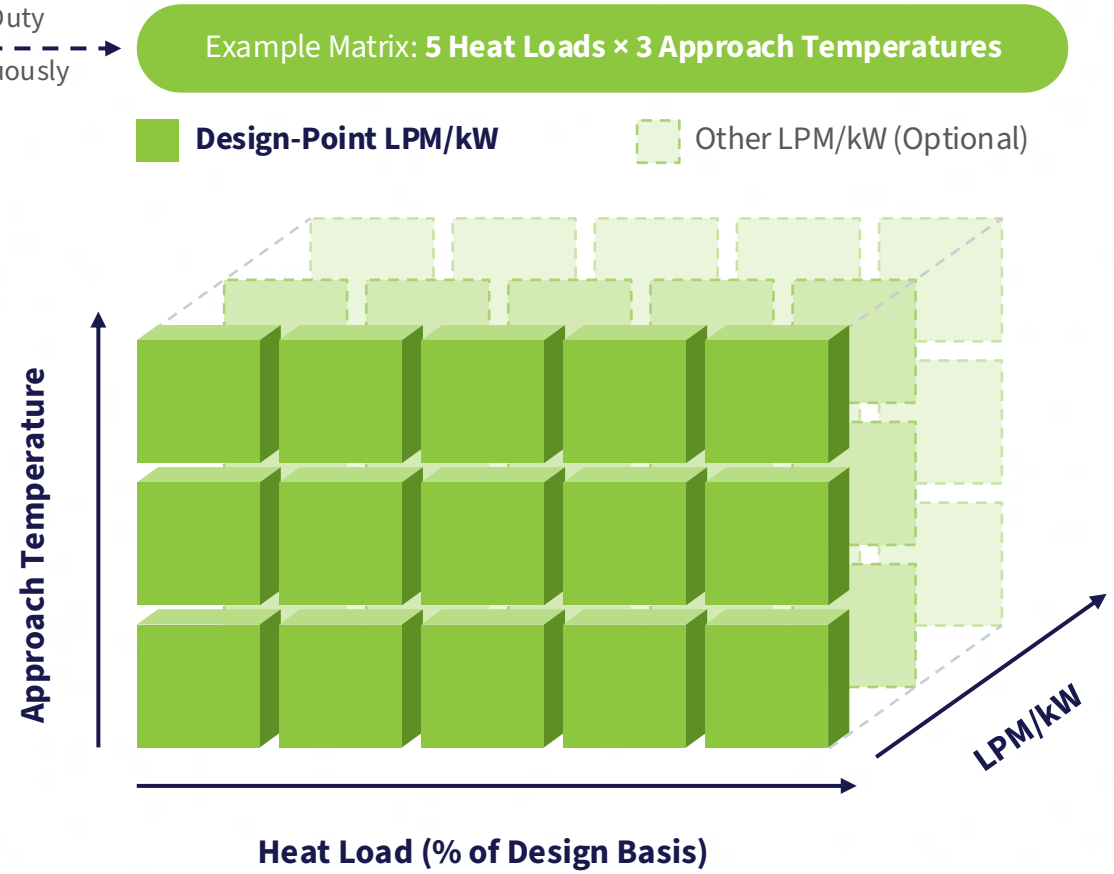
This is done for both Copper tube and AL tube HX and the cooling power is similar Cooling evaluation:

	cooling power (kW)	Air pressure drop (Pa)	liquid pressure drop (kPa)
Flat HX	176kW	420	250
slant HX	177kW	390	260
V shape HX	179kW	150	300

Test Setup & Steady-State Test Matrix

A common test setup and test matrix let operators compare AALC vendors and plan data center capacity

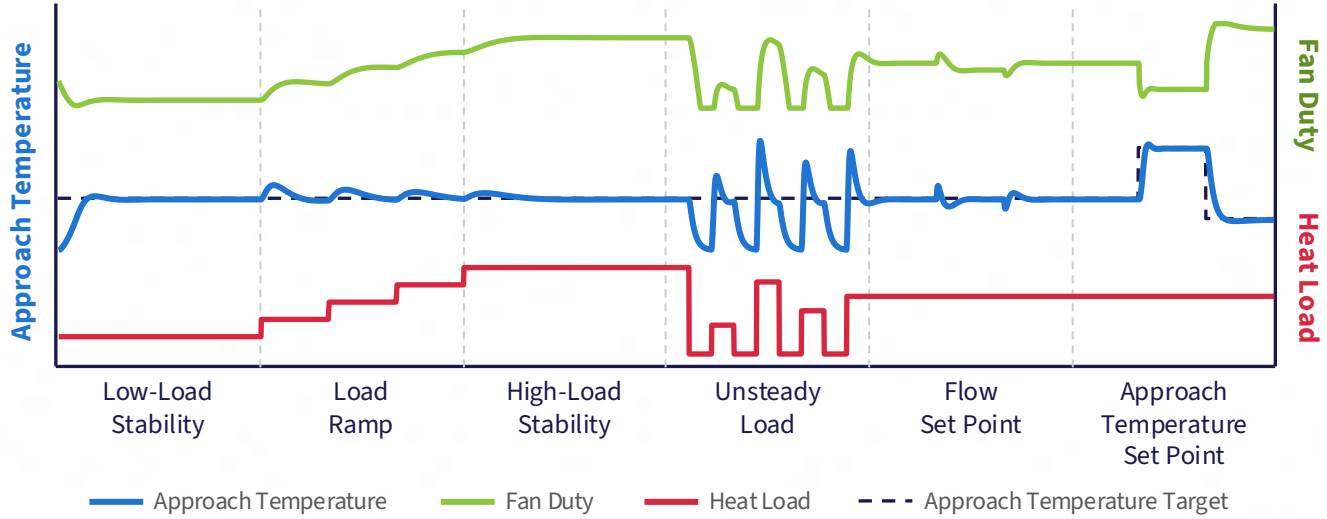




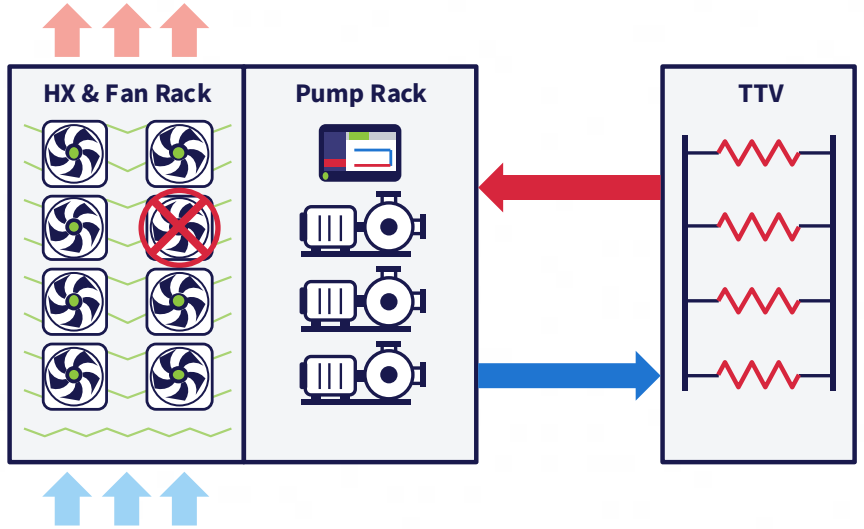
Transient, Redundancy & Supporting Tests

Verify controller transient response and component redundancy, and gather supporting data

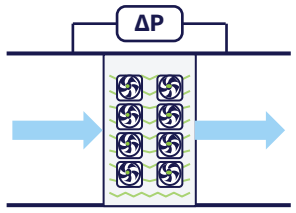
Controller Response



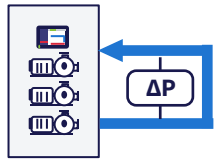
Redundancy



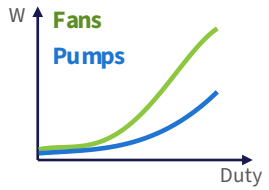
Supporting Characterization



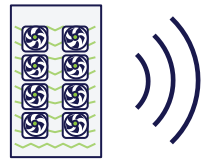
Air-Side P-Q



Liquid-Side P-Q



Fan & Pump Power



Acoustics

Fail a Redundant Component (e.g., a Fan)

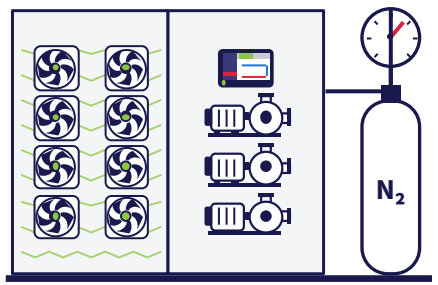
Still Holds Design-Basis Approach Temperature at Design Heat Load

Logistics & Serviceability

Plan the path from factory to data hall, and design for service in the row

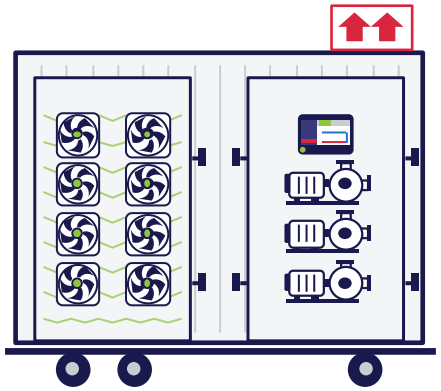
Ship & Commission

Factory



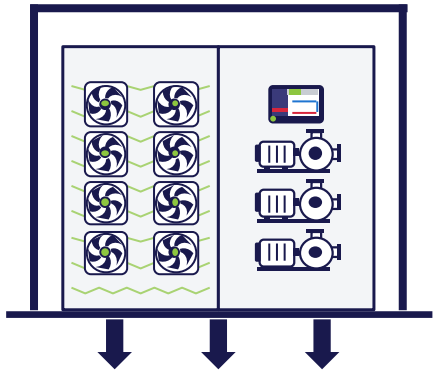
Leak Test & Functional Test

Transit



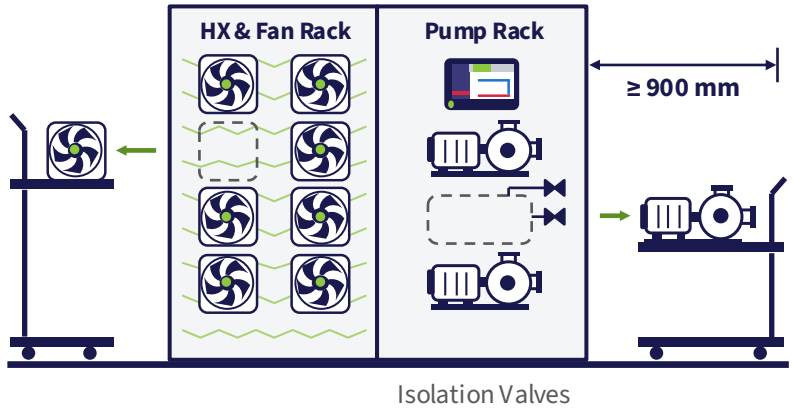
Ship Upright, Split to Fit

Site



Check Clearances & Floor Load

Service in the Row

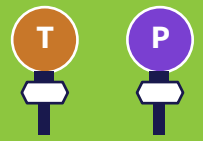


Hot-Swap from the Aisle

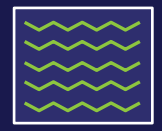
Design Around Component Criticality



Critical
Redundant & Hot-Swap

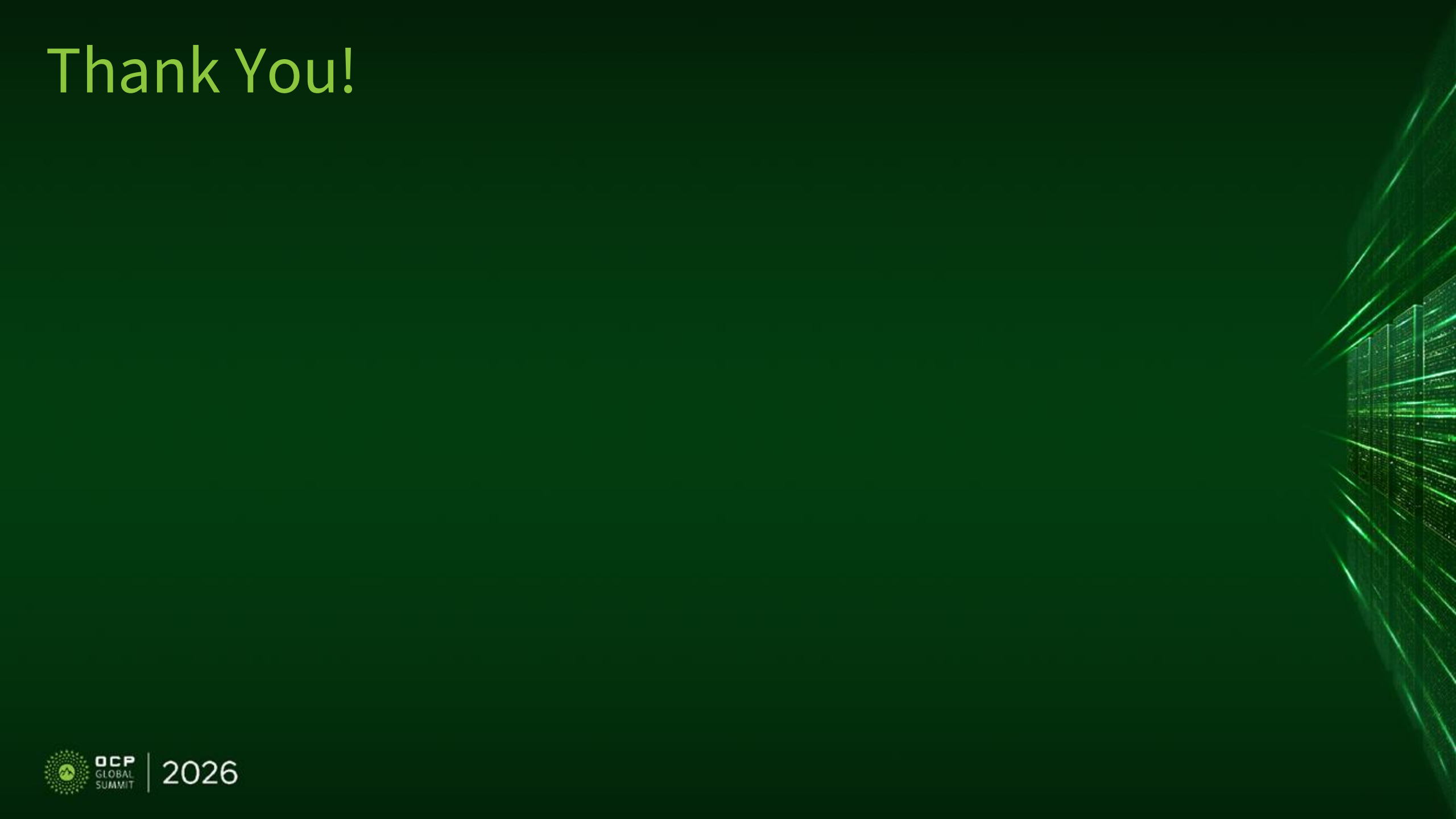


Non-Critical
Field-Replaceable



Single Point of Failure
Design for Unit Life

Thank You!



OCP Base Specification – Door Heat Exchanger

Ashok Yadav *Global Offering Manager HDLC, VERTIV*

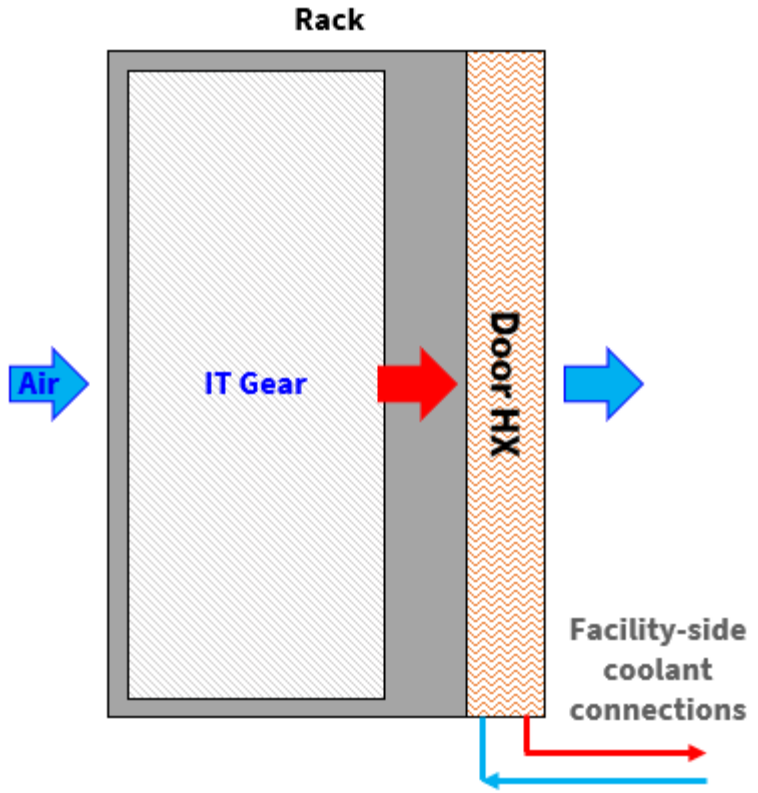
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Open Rack Door Heat Exchangers: A Common Path to Higher-Density Air Cooling

Core message: This specification provides practical interoperability guidance for integrating rear-mounted Door Heat Exchangers (Door HX) with Open Rack systems, enabling a common, vendor-neutral path to higher-density cooling.

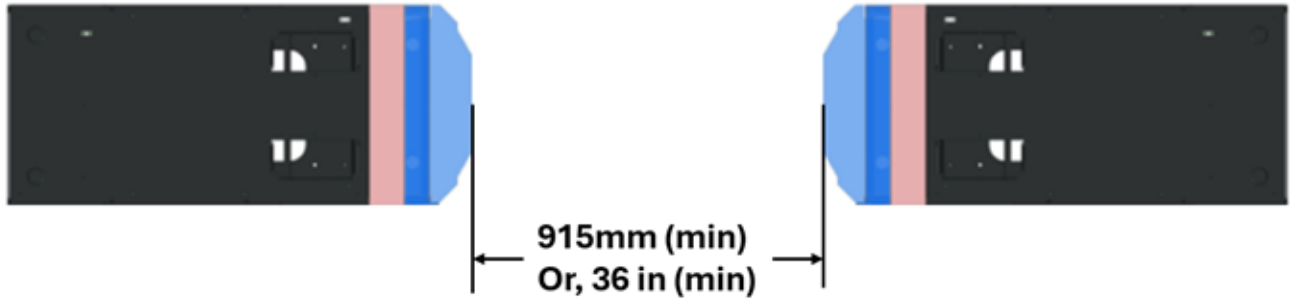
- Meets the cooling demands of the AI era, where surging rack densities have outpaced what conventional air cooling alone can deliver
- Defines a rear-mounted Door HX reference architecture aligned to Open Rack V3
- Supports both passive and active Door HX designs across three deployment models:
    - Air-to-liquid rear-door cooling for traditional air-cooled IT
    - Hybrid racks combining cold plates (CPUs/GPUs) with Door HX for residual air heat
    - Liquid-to-air systems for facilities that retain room-air cooling
- Delivers consistent deployment, serviceability, and thermal performance across vendors and sites



What the New Specification Defines

Core message: This specification establishes a shared engineering baseline so that operators and vendors can deploy Door Heat Exchangers at AI-scale density with confidence, consistency, and

Area	What the Specification Establishes
Mechanical Fit	Ensures Door HX units mount seamlessly onto Open Rack V3 frames, with a maximum depth of 305 mm for passive and 370 mm for active designs to preserve aisle space
Facility Layout	Requires a minimum 915 mm (36 in) aisle clearance between closed doors so technicians can service racks safely without disrupting adjacent rows
Structural Limits	Caps preferred Door HX weight at 100 kg for standard-density racks, allows up to 150 kg with additional support, and defines a structured evaluation path for extreme-density doors up to 200 kg, including guidance on suitable rack configurations for such deployments
Fluid & Power	Standardizes a single coolant supply/return loop with OCP-compatible quick-disconnect fittings where feasible, supporting both AC and 48 VDC power configurations
Thermal Reporting	Creates a universal performance benchmark: all vendors rate capacity at 17 °C coolant inlet and 45 °C IT exhaust, enabling apples-to-apples comparison across products, with server inlet temperature guidelines aligned to ASHRAE TC9.9 (18 °C–27 °C)
Efficiency Targets	Sets measurable efficiency goals: passive doors must stay below 15 Pa air-side pressure drop; active doors must consume less than 2% of their rated cooling capacity in fan power
Operations & Management	Provides a complete operational framework covering real-time monitoring, alarm thresholds, leak testing procedures, field serviceability, and Redfish-aligned remote management



Where the Spec Fell Short

Core message: Revision 1.0 establishes a strong baseline but deliberately leaves several areas for community validation and future normative work.

- **Open Rack Wide (ORW)**: Recognized, but detailed rear-door requirements are deferred to a future chapter or revision
- **Deployment-specific performance**: The spec permits manufacturers to provide full capacity, impedance, and power curves on request rather than requiring a universal public reporting package
- **High-density structural qualification**: Doors above legacy weight ranges require documented rack, transition-frame, hinge-load, center-of-gravity, and floor-loading analysis, but a common validation template is still needed
- **Aisle interaction**: CFD or equivalent review is recommended for opposing-row and multi-door deployments; standardized scenarios and acceptance criteria are not yet fully defined
- **Management interoperability**: Redfish alignment is encouraged “where feasible,” leaving room for a more prescriptive schema, telemetry set, and control-model definition
- **Safety and test harmonization**: The document permits alternative recognized test methods and safety bases; stronger cross-vendor test comparability would improve procurement confidence

[Call to Action]

Core message: The next revision should be driven by operators, rack vendors, Door HX suppliers, IT OEMs, and facility teams with real deployment data.

- **Operators**: Share aisle layouts, rack-density ranges, coolant conditions, maintenance constraints, and failure-mode requirements
- **Door HX and rack suppliers**: Validate fit, door loading, blind-mate fluid interfaces, airflow sealing, and service workflows across implementations
- **IT and platform teams**: Contribute server airflow/backpressure limits and hybrid-cooling operating profiles
- **Management contributors**: Define interoperable Redfish telemetry, alarms, controls, and service data models
- **Thermal engineers**: Develop common CFD cases and test/acceptance methods for high-density and opposing-row installations
- **Workgroup objective**: Advance ORW requirements, standardized performance reporting, degraded-mode characterization, and field-proven reliability requirements

Thank You!

